Smart Campus Canteen Pre-Ordering & Live Token System

Full-Stack Web Engineering Project • Java (Spring Boot) & React

PROJECT ABSTRACT

The Smart Campus Canteen Pre-Ordering & Live Token System eliminates long lunch queues and chaotic food counters in college campuses through an intuitive full-stack web application. Built with **React** and **Spring Boot (Java)**, the platform enables students to pre-order meals from dynamic menus and receive instant digital tokens for scheduled pickups. Canteen staff manage incoming orders through a real-time kitchen queue, while a public TV display board flashes ready tokens for hassle-free collection. Secured with **JWT authentication** and powered by **Spring Data JPA**, this project modernizes campus dining into a fast, transparent, and completely queue-free experience.

Key System Modules

1. Student Portal (React) • Browse categorized daily menus (Breakfast, Lunch, Snacks, Drinks). • Add to interactive cart with instant price calculation. • Select class-break pickup time slots. • Real-time token tracking badge (Placed → Preparing → Ready).	**2. Canteen Staff Dashboard (Spring Boot + React)** • Live incoming kitchen order queue ordered by pickup time. • 1-click status transitions (Received → Preparing → Ready). • Live item availability toggle (Mark 'Out of Stock' instantly). • Daily order summary and sales breakdown.
3. Live Public Display Board • Fullscreen high-visibility screen for dining hall TV/monitor. • Clear status split: *Now Preparing* vs *Ready for Pickup*. • Color-coded flashing alerts for recently ready tokens. • Eliminates counter crowding and verbal announcements.	**4. Security & Data Management** • Stateless authentication via **JSON Web Tokens (JWT)**. • Role-Based Access Control (ROLE_STUDENT & ROLE_STAFF). • Relational database integrity with Spring Data JPA & Hibernate. • Prevents duplicate token collisions and race conditions.

Technology Specifications

Layer	Technology Choices	Functional Role
Frontend	React 18+, Vite, Tailwind CSS, Axios	Interactive UI, responsive cart state, live status polling
Backend	Java 17/21, Spring Boot 3.x, Spring Web	RESTful endpoints, business logic, token generation
Security	Spring Security 6, JWT, BCrypt	Stateless token auth, role authorization (Student/Staff)
Database	Spring Data JPA, Hibernate, MySQL/PostgreSQL	Relational data mapping, ACID transactions, data persistence
Deployment	Docker, GitHub / Localhost	Containerized multi-container full-stack execution